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COLLEGE OF COMPUTER, INFORMATION AND COMMUNICATIONS TECHNOLOGY

PC 317
Systems Analysis and Design

Worksheet No. 5: Economic Feasibility Analysis		Date:
Student Names:	1. 2. 3. 4.	

Objectives : Given the materials, tools and equipment, upon completion of the activity, the student must be able to:

1. Analyze the economic feasibility of the proposed project.
2. Develop a cost-benefit analysis.

Background :

Economic Feasibility

The second element of a feasibility analysis is to perform an *economic feasibility* analysis (also called a *cost-benefit analysis*). This attempts to answer the question “Should we build the system?” Economic feasibility is determined by identifying costs and benefits associated with the system, assigning values to them, calculating future cash flows, and measuring the financial worthiness of the project.

As a result of this analysis, the financial opportunities and risks of the project can be understood. Keep in mind that organizations have limited capital resources and multiple projects will be competing for funding. The more expensive the project, the more rigorous and detailed the analysis should be. Before illustrating this process with a detailed example, we first introduce the framework we will apply to evaluate project investments and the common assessment measures that are used.

Economic Feasibility: Should We Build It?

- Development costs
- Annual operating costs
- Annual benefits (cost savings and/or increased revenues)
- Intangible benefits and costs

Figure in the next page shows the steps involved in performing an economic feasibility analysis. Each step will be illustrated by an example in the upcoming sections.

1. Identify Costs and Benefits	List the tangible costs and benefits for the project. Include both one-time and recurring costs.
2. Assign Values to Costs and Benefits	Work with business users and IT professionals to create numbers for each of the costs and benefits. Even intangibles should be valued if at all possible.
3. Determine Cash Flow	Forecast what the costs and benefits will be over a certain period, usually, three to five years. Apply a growth rate to the values, if necessary.
4. Assess Project's Economic Value	Evaluate the project's expected returns in comparison to its costs. Use one or more of the following evaluation techniques:
• Return on Investment (ROI)	Calculate the rate of return earned on the money invested in the project, using the ROI formula.
• Break-Even Point (BEP)	Find the year in which the cumulative project benefits exceed cumulative project costs. Apply the breakeven formula, using figures for that year. This calculation measures how long it will take for the system to produce benefits that cover its costs.
• Net Present Value (NPV)	Restate all costs and benefits in today's dollar terms (present value), using an appropriate discount rate. Determine whether the total present value of benefits is greater than or less than the total present value of costs.

1. Identify Costs and Benefits

The *systems analyst's first task when developing an economic feasibility analysis is to identify the kinds of costs and benefits the system will have* and list them along the left-hand column of a spreadsheet. The costs and benefits can be broken down into four categories: (1) development costs, (2) operational costs, (3) tangible benefits, and (4) intangibles.

Development Costs	Operational Costs
Development team salaries	Software upgrades
Consultant fees	Software licensing fees
Development training	Hardware repairs
Hardware and software	Hardware upgrades
Vendor installation	Operational team salaries
Office space and equipment	Communications charges
Data conversion costs	User training
Tangible Benefits	Intangible Benefits
Increased sales	Increased market share
Reductions in staff	Increased brand recognition
Reductions in inventory	Higher quality products
Reductions in IT costs	Improved customer service
Better supplier prices	Better supplier relations

Development costs are those tangible expenses that are incurred during the creation of the system, such as salaries for the project team, hardware and software expenses, consultant fees, training, and office space and equipment. Development costs are usually thought of as one-time costs.

Operational costs are those tangible costs that are required to operate the system, such as the salaries for operations staff, software licensing fees, equipment upgrades, and communications charges. Operational costs are usually thought of as ongoing costs.

Tangible benefits include revenue that the system enables the organization to collect, such as increased sales. In addition, the system may enable the organization to avoid certain costs, leading to another type of tangible benefit: cost savings. For example, if the system produces a reduction in needed staff, lower salary costs result. Similarly, a reduction in required inventory levels due to the new system produces lower inventory costs. In these examples, the reduction in costs is a tangible benefit of the new system.

Of course, a project also can affect the organization's bottom line by reaping *intangible benefits* or incurring *intangible costs*. Intangible costs and benefits are more difficult to incorporate into the economic feasibility analysis because they are based on intuition and belief rather than on "hard numbers." Nonetheless, they should be listed in the spreadsheet along with the tangible items.

2. Assign Values to Costs and Benefits Once the types of costs and benefits have been identified, the analyst needs to assign specific dollar values to them. This may seem impossible—How can someone quantify costs and benefits that haven't happened yet? And how can those predictions be realistic? Although this task is very difficult, you have to do the best you can to come up with reasonable numbers for all of the costs and benefits. Only then can the approval committee make an informed decision about whether or not to move ahead with the project.

The most effective strategy for estimating costs and benefits is to rely on the people who have the best understanding of them. For example, costs and benefits that are related to the technology or the project itself can be provided by the company's IT group or external consultants, and business users can develop the numbers associated with the business (e.g., sales projections, order levels). The company also can consider past projects, industry reports, and vendor information, although these sources probably will be a bit less accurate. Likely, all of the estimates will be revised as the project proceeds.

If predicting a specific value for a cost or benefit is proving difficult, it may be useful to estimate a range of values for the cost or benefit and then assign a likelihood (probability) estimate to each value. With this information, an *expected value* for the cost or benefit can be calculated. As more information is learned during the project, the value estimates and the probability estimates can be revised, resulting in a revised expected value for the cost or benefit.

What about the *intangible* benefits and costs? Sometimes, it is acceptable to list intangible benefits, such as improved customer service, without assigning a dollar value. Other times, estimates have to be made regarding how much an intangible benefit is "worth." We suggest that you quantify intangible costs or benefits if at all possible. If you do not, how will you know if they have been realized? Suppose that a system claims to improve customer service. This benefit is intangible, but let's assume that the improvement in customer service will decrease the number of customer complaints by 10% each year over three years and that \$200,000 is currently spent on phone charges and phone operators who handle complaint calls. Suddenly, we have some very tangible numbers with which to set goals and measure the originally intangible benefit.

3. Determine Cash Flow A formal cost-benefit analysis usually contains costs and benefits over a selected number of years (usually, three to five years) to show cash flow over time. Same amount for customer complaint calls, inventory costs, hardware, and software for all four years. Often, amounts are augmented by some rate of growth to adjust for inflation or business improvements, as shown by the 6% increase that is added to the sales numbers in the sample spreadsheet. Similarly, labor costs are assumed to increase at a 4% rate each year. Finally, totals are added to determine what the overall benefits and costs.

4. Assist Project's Economic Value

Return on Investment The return on investment (ROI) is a calculation that measures the average rate of return earned on the money invested in the project. ROI is a simple calculation that divides the project's net benefits (total benefits _ total costs) by the total costs. The ROI formula is:

$$\text{ROI} = \frac{\text{Total Benefits} - \text{Total Costs}}{\text{Total Costs}}$$

$$\text{ROI} = \frac{152,000 - 138,000}{138,000} = \frac{14,000}{138,000} = 10.14\%$$

A high ROI suggests that the project's benefits far outweigh the project's cost, although exactly what constitutes a "high" ROI is unclear. ROI is commonly used in practice; however, it is hard to interpret and should not be used as the only measure of a project's worth.

Break-Even Point Another common approach to measuring a project's worth is the break-even point. The *break-even point* (also called the *payback method*) is defined as the number of years it takes a firm to recover its original investment in the project from net cash flows.

(In the year in which Cumulative Cash Flow turns positive):

$$\text{BEP} = \text{Number of years of negative cash flow} + \frac{\text{That year's Net Cash Flow} - \text{That year's Cumulative Cash Flow}}{\text{That year's Net Cash Flow}}$$

Using the values in Figure 1-8, the BEP calculation is:

$$\text{BEP} = 2 + \frac{41,000 - 14,000}{41,000} = 2 + \frac{28,000}{41,000} = 2.68 \text{ years}$$

The basic formula to convert a future cash flow to its present value is:

$$\text{PV} = \frac{\text{Cash flow amount}}{(1 + \text{rate of return})^n} \quad \text{where } n \text{ is the year in which the cash flow occurs.}$$

The rate of return used in the present value calculation is sometimes called the required rate of return, or the cost of obtaining the capital needed to fund the project. Many organizations will have determined the appropriate rate of return to use when analyzing IT investments. The systems analyst should consult with the organization's finance department.

Net Present Value (NPV) The NPV is simply the difference between the total present value of the benefits and the total present value of the costs.

$$\text{NPV} = \sum \text{PV of Total Benefits} - \sum \text{PV of Total Costs}$$

$$= \$125,056 - \$131,029 = (\$5,973)$$

As long as the NPV is greater than zero, the project is considered economically acceptable. Unfortunately for this project, the NPV is less than zero, indicating that for a required rate of return of 10%, this project should not be accepted. The required rate of return would have to be something less than 6.65% before this project returns a positive NPV. This example illustrates the fact that sometimes the "naïve" techniques of ROI and BEP find that the project appears acceptable, but the more rigorous and financially correct NPV technique finds the project is actually unacceptable.

Resources: System Analysis & Design 5th Edition by Dennis, Wixom, and Roth pages 25-32"

Accomplish the following task:

1. Create an economic feasibility analysis of the proposed system.
(refer to the given sample below as guide)

Economic Feasibility

A cost-benefit analysis was performed; see attached spreadsheet for details (provided in Appendix 1A). Conservative estimates show that the Digital Music Download system has a good chance of significantly enhancing the company's bottom line.

ROI over 3 years: 280%

NPV over 3 years: \$4,180,431

Break-even occurs after 0.17 years

Intangible Costs and Benefits

Improved customer satisfaction.

Enhanced competitive position through expansion of our brand into the music download market.

	2012	2013	2014	2015	Total
Benefits					
Increased sales from individual music downloads		757,500	787,800	811,434	2,356,734
Increased sales from customer subscriptions		950,000	988,000	1,017,640	2,955,640
Increased sales from in-store or website CD sales		205,000	213,200	219,596	637,796
Increased sales from music download gift cards		153,000	159,120	163,894	476,014
Total Benefits		2,065,500	2,148,120	2,212,564	6,426,184
Present Value Total Benefits		1,948,585	1,911,819	1,857,711	5,718,115
Development Costs					
Labor: Analysts and design	42,000	0	0	0	42,000
Labor: Implementation	120,000	0	0	0	120,000
Consultant fees	50,000	0	0	0	50,000
Development training	5,000	0	0	0	5,000
Office space and equipment	2,000	0	0	0	2,000
In-store kiosks	25,000	0	0	0	25,000
Software	10,000	0	0	0	10,000
Hardware	25,000	0	0	0	25,000
Total Development Costs	279,000	0	0	0	279,000
Operational Costs					
Labor: Webmaster		85,000	87,550	90,177	262,727
Labor: Network technician		60,000	61,800	63,654	185,454
Labor: Computer operations		50,000	51,500	53,045	154,545
Labor: Business manager		60,000	61,800	63,654	185,454
Labor: Assistant manager		45,000	46,350	47,741	139,091
Labor: Three staff		90,000	92,700	95,481	278,181
Software upgrades		1,000	1,000	1,000	3,000
Software licenses		3,000	1,000	1,000	5,000
Hardware upgrades		5,000	3,000	3,000	11,000
User training		2,000	1,000	1,000	4,000
Additional ISP charges		15,000	17,000	18,500	50,500
Communications charges		20,000	20,000	20,000	60,000
Marketing expenses		25,000	25,000	25,000	75,000
Total Operational Costs		461,000	469,700	483,251	1,413,951
Total Costs	279,000	461,000	469,700	483,251	1,692,951
Total Benefits – Total Costs	(279,000)	1,604,500	1,678,420	1,729,313	4,733,233
Cumulative Net Cash Flow	(279,000)	1,325,500	3,003,920	4,733,233	
Present Value Total Costs	279,000	434,906	418,031	405,747	1,537,684
Return on Investment (ROI)	280%	(6,426,184/1,692,951)			
Break-Even Point	0.17 years	(costs are fully recovered in the first year; [1,604,500 – 1,325,500]/1,604,500)			
NPV (PV Total Benefits – PV Total Costs)	4,180,431	(5,718,115 – 1,537,684)			
Intangible Benefits:	Improved customer satisfaction Enhanced market position				